

Uwaise Ibna Islam

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EDUCATION

University of Florida PhD in Computer Science (Bioinformatics) Advisor: Dr. Christina Boucher · Research: scalable algorithms for population genomics, variant analysis, genomic data structures	Expected May 2028 GPA: 4.00
Florida Atlantic University MS in Computer Science Presidential Fellowship; Academic Excellence Award (2023, 2024)	Aug 2022 – May 2024 GPA: 3.97
Chittagong University of Engineering & Technology BS in Computer Science & Engineering	Feb 2016 – Jun 2021

TECHNICAL SKILLS

Languages	Python, R, C++, Bash/Shell, SQL
NGS & Bioinformatics	WGS/WES variant calling, VCF/BCF, FASTA/FASTQ, BAM/CRAM, HTSlib, bcftools, samtools, GATK, PLINK, SHAPEIT, BEDtools, Bioconductor, sequence alignment, phylogenetics
Pipelines & Infra	Nextflow, Snakemake, SLURM/HPC, Docker, CI/CD, Google Cloud Platform, AWS
ML & Data Science	PyTorch, TensorFlow, scikit-learn, deep learning, feature engineering, NumPy, pandas
Systems	Linux/UNIX, Git/GitHub, multithreading (OpenMP), memory profiling (perf, Valgrind)

RESEARCH EXPERIENCE

Graduate Research Assistant · University of Florida, Boucher Lab	Jul 2025 – Present
- Designed and implemented PBML, a C++17 tool for scalable haplotype matching on WGS panels from the 1000 Genomes Project (5,008 haplotypes, 1–6M variants/chr) and BIG Initiative biobank (11,000 haplotypes, 893K sites). - Achieved 5× faster queries than μ-PBWT and 2.6× faster than Durbin's PBWT with 44× lower peak memory (4 GB vs 176 GB), using SDSL compressed bitvectors and HTSlib for VCF/BCF I/O. Scaled to 49K haplotypes while staying 2.2× faster with 2.2× less memory. - Built OpenMP multi-threaded query engine: 8.2× speedup on 16 threads with 0.6% memory overhead, vs. 15.9% growth in competing tools. Length-threshold filtering reduced query time by 74% and output by 98% while retaining 97% coverage—enabling practical IBD detection at population scale.	
Graduate Research Assistant · Florida Atlantic University, Assis Group	Aug 2022 – May 2025
- Developed end-to-end deep learning pipeline (PyTorch) for cross-species genotype–phenotype prediction with custom feature engineering on multi-species genomic alignments (59 species). Near-perfect cross-validated accuracy across multiple phenotype tasks. - Authored R/Bioconductor package with parallel genomic analysis workflows and integrated neural network inference; achieved 7× speedup through optimized preprocessing. Published in <i>Briefings in Bioinformatics</i> (2025).	

PROJECTS

HPC Genomic Pipelines	Python, Bash, Nextflow, SLURM, Docker
Built automated, reproducible workflows for phasing and variant calling on large-scale WGS datasets (SHAPEIT5, Beagle) on HPC clusters and cloud. Designed job scheduling, containerized environments, and error recovery for batch processing across 22 chromosomes.	

PUBLICATIONS

- Islam U.I.**, Cozzi D., Gagie T., Bonizzoni P., Prezza N., Rossi M. Scalable PBWT Queries with Minimum-Length SMEM Constraints. Submitted to *RECOMB 2026* (under review).
- Islam U.I.**, Campelo dos Santos A.L., Kanjilal R., Assis R. Learning genotype-phenotype associations from gaps in multi-species sequence alignments. *Briefings in Bioinformatics*, 2025.

INDUSTRY EXPERIENCE

Software QA Engineer · Enosis Solutions	Aug 2021 – Jul 2022
Automated end-to-end testing with Python/Selenium (63% coverage, 40% fewer regression bugs). Performed API performance testing and collaborated cross-functionally in agile sprints to achieve 82% response time improvement.	

HONORS & CERTIFICATIONS

FAU Presidential Fellowship (2022–2024) · Academic Excellence Award (2023, 2024) · IBM Data Science Professional Certificate · ML Specialization (Coursera)